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Introduction to p-adic Numbers
Homework Sheet 2

Due: October 28

Recall the definitions from the lecture. A *valuation* $v\colon R \to \mathbb{R} \cup \{\infty\}$ on a commutative ring R is a function satisfying the following properties:

V(i) $v(x) = \infty$ iff $x = 0$;

V(ii) $v(xy) = v(x) + v(y)$ for all $x, y \in R$;

V(iii) $v(x + y) \geq \min\{v(x), v(y)\}$.

A *norm* $|-|\colon R \to \mathbb{R}_{\geq 0} \cup \{\infty\}$ is a function satisfying the following properties:

N(i) $|x| = 0$ iff $x = 0$;

N(ii) $|xy| = |x| \cdot |y|$ for all $x, y \in R$;

N(iii) $|x + y| \leq |x| + |y|$ for all $x, y \in R$.

A norm $|-|$ is called *nonarchimedean* if it satisfies

N(iv) $|x + y| \leq \max\{|x|, |y|\}$ for all $x, y \in R$.

Otherwise, the norm $|-|$ is called *archimedean*.

Exercise 2.1

Valuations and norms. For a commutative integral domain R let K be the field of fractions for R. (Integral domain and field of fractions are explained in the script, see Bemerkung 2.9.)

(a) Given any valuation $v\colon R \to \mathbb{R} \cup \{\infty\}$, extend it to K uniquely. Prove that it is a valuation on K.

(b) Given any nonarchimedean norm $|-|\colon R \to \mathbb{R}_{\geq 0} \cup \{\infty\}$, extend it to K uniquely. Prove that it is a nonarchimedean norm on K.

(c) Given any valuation $v\colon R \to \mathbb{R} \cup \{\infty\}$, extend it to K uniquely and let $|-|_v\colon K \to \mathbb{R}_{\geq 0}$, $|x|_v := e^{-v(x)}$ for all $x \in K$. Prove that $|-|_v$ is a nonarchimedean norm on K. Conversely, for any nonarchimedean norm $|-|\colon K \to \mathbb{R}_{\geq 0} \cup \{\infty\}$, $-\log|-|\colon K \to \mathbb{R} \cup \{\infty\}$ is a valuation on K.

Exercise 2.2

Nonarchimedean norm on $\mathbb{Q}$. For a prime p the p-adic valuation $v_p\colon \mathbb{Q} \to \mathbb{R} \cup \{\infty\}$ is the unique extension of the function measuring the maximal divisibility of a nonzero integral number by p. Let $v_p(0) := \infty$. The p-adic norm is $|x|_p := p^{-v_p(x)}$. Prove that $|-| := |-|_p$ is a nonarchimedean norm.

Exercise 2.3

Formulas for valuations on $\mathbb{Q}$.

(a) Find $v_5(-0.0625)$.

(b) $v_p((p^N)!) = 1 + p + p^2 + \cdots + p^{N-1}$ for any prime p and any integer N.

Exercise 2.4

Properties of norms on $\mathbb{Q}$. For any positive rational $x \in \mathbb{Q}$, $\prod_p |x|_p = 1/x$, where p runs over all prime numbers.